

Research paper

An improved solar step-up power converter for next-generation electric vehicle charging

Hwa-Dong Liu^{*}, Wen-Tung Huang

Undergraduate Program of Vehicle and Energy Engineering, National Taiwan Normal University, Taipei 106, Taiwan, ROC

ARTICLE INFO

Keywords:

Solar step-up power converter
Pyramid maximum power point tracking
Electric vehicle charging
Discontinuous conduction mode

ABSTRACT

This study proposes an innovative control strategy based on a quadratic equation derived from a core battery charging model. This strategy is applied to a solar step-up power converter (SSUPC), which is specifically optimized for electric vehicle charging. The model includes a 500 W SSUPC, controlled by a microprocessor, effectively converting low input voltage into high output voltage. The proposed control strategy allows the SSUPC to operate in discontinuous conduction mode, thereby achieving a higher voltage gain. In experiments with an input voltage of 20 V, the system was capable of outputting up to 480 V, while maintaining a low duty cycle of 0.3 and achieving a system conversion efficiency of 95 %. Furthermore, the SSUPC integrates a pyramid maximum power point tracking (MPPT) algorithm, initiating MPPT to capture the maximum power point when the solar output power is insufficient for electric vehicle charging, achieving an MPPT efficiency of 99 %. The developed charger is compact, boasts high power density, is cost-effective, and is well-suited for widespread application, meeting the 480 V charging requirements of electric vehicles.

1. Introduction

The emergence of electric vehicles (EVs) dates back to the 19th century, beginning with Anderson's invention of the first electric car, which ran on a disposable battery. This era witnessed the rise of rechargeable battery technology, paving the way for rechargeable electric vehicles. However, the early 20th century saw a decline in EV development, overshadowed by significant advancements in internal combustion engine technology and oil extraction (Li et al., 2017). Renewed interest in electric vehicle technology emerged in the late 20th century, driven by increasing concerns over environmental pollution and the energy impact of gasoline-powered vehicles, leading to the widespread presence of EVs on roads today (Thangavel et al., 2023). Presently, most EVs are charged at charging stations, which are categorized into two types based on the charging method: alternating current (AC) and direct current (DC) charging. AC charging transfers AC electricity from the station to the EV through a cable, which is then converted to DC by an onboard charger to power the battery (Karneddi and Ronanki, 2022). Conversely, DC charging involves converting electricity to DC within the charging station itself and directly charging the EV (Rathore et al., 2022). AC charging is generally less powerful than DC charging due to the limitations imposed by the size of the

onboard charger, while DC stations can include multiple boost modules for higher output power, resulting in faster charging times. However, DC charging stations are more expensive because of their complex AC/DC converter structure and the design of their charging modules (Shahverdi et al., 2023; Ramanathan et al., 2024; Hakam et al., 2024; Chen et al., 2023; Kumar et al., 2023; Deshmukh et al., 2022).

Since the advent of DC charging technology for electric vehicles, boost converters have undergone a variety of design and functional enhancements (Sayed and Massoud, 2022). Recent years have witnessed a surge in innovative designs. For example, in 2019, Emamalipour and colleagues unveiled a bidirectional DC/DC converter that incorporates a hybrid inverter-rectifier structure with a capacitor-inductor-inductor-capacitor (CLLC) resonant circuit. This design utilizes the CLLC resonant circuit for voltage elevation and operates in rectifier mode to minimize voltage ripples, requiring fewer high-voltage switches than traditional dual active bridge converters (Emamalipour and Lam, 2020). In 2020, Wang et al. introduced an integrated multi-stage boost converter that effectively regulates power flow over a broad load range, eliminating the need for secondary-side regulators (Wang and Thrimalithana, 2020). Furthermore, in 2021, Mamun and team developed a DC converter that combines cascaded boost and switched-capacitor zeta (SC-ZETA) topologies, achieving exceptionally high voltage gain

^{*} Corresponding author.

E-mail addresses: hdliu@ntnu.edu.tw (H.-D. Liu), wentong0901@gmail.com (W.-T. Huang).

<https://doi.org/10.1016/j.egy.2024.08.014>

Received 2 June 2024; Received in revised form 10 July 2024; Accepted 5 August 2024

Available online 8 August 2024

2352-4847/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

suitable for high-power applications (Mamun et al., 2021). In 2023, Sharma et al. introduced a design for a bidirectional onboard charger, specifically tailored for light electric vehicles. Distinguished by its multistep constant-current charging capability, this charger excels in both high performance and cost efficiency (Sharma and Singh, 2023a). In 2023, Sharma et al. introduced a battery charger with bidirectional operability, designed for low-voltage, battery-powered vehicles. A notable advantage of this charger lies in its effective performance, which extends across a wide output voltage range (Sharma and Singh, 2023b). In 2023, Rafi et al. highlighted the semi-dual active bridge converter as an ideal option for vehicle chargers. This preference is attributed to its adaptability across a wide range of input and output voltages, coupled with its proven high performance (Rafi and Bauman, 2023). In 2023, Zheng et al. proposed a novel bidirectional inductor-inductor-capacitor plus capacitor (LLC+C) converter as a solution to this challenge. It operates as a full-bridge inductor-inductor-capacitor (LLC) converter in the grid-to-vehicle (G2V) direction and transitions to a half-bridge inductor-inductor-capacitor-capacitor (LLCC) converter with a smaller transformer in the vehicle-to-grid (V2G) direction. This innovative design adeptly meets specific operational requirements while also delivering robust system performance (Zheng and Bauman, 2023). In 2023, Lenka et al. introduced an innovative control algorithm for a grid-interactive off-board EV battery charger. This algorithm, featuring a self-tuning filter and sliding mode control, is designed not only to efficiently power EV batteries but also to simultaneously enhance the power quality of the grid (Lenka et al., 2023). In 2023, Nazi et al. introduced a single-phase, bridgeless, electrolytic capacitor-less, single-stage battery charger. This charger, featuring a bridgeless power factor correction circuit and an isolated series resonant converter, is particularly notable for its low energy loss and high stability (Nazi et al., 2023). In 2023, Payarou et al. explored the integrated multipurpose power electronics interface, a device versatile enough for use in vehicle chargers. Capable of facilitating both V2G and G2V operations, this device notably enhances grid flexibility (Payarou and Pillay, 2023).

Additionally, solar power technology has attracted many researchers to develop maximum power point tracking (MPPT) techniques (Kong et al., 2024; Wesabi et al., 2024; Naamane et al., 2024; Tia et al., 2024). In 2024, Kong and others proposed an improved snake optimizer algorithm for MPPT control, which not only shortens the convergence time of the traditional snake optimizer algorithm but also enhances the accuracy of locating the maximum power point, thereby boosting the overall system efficiency (Kong et al., 2024). In 2024, Wesabi and colleagues conducted a research study on improvements to the incremental conductance technique, taking into full consideration the effects of solar irradiance and load. They flexibly adjusted the duty cycle step size, aiming to enhance the output power and voltage of solar panels while reducing switching and conduction losses, thereby boosting the overall system performance (Wesabi et al., 2024). In 2024, Naamane et al. developed a technology for photovoltaic (PV) system management that integrates distributed power resources with a multifunctional voltage source inverter to regulate the maximum power point (MPP) of the PV system. This method aims to enhance the efficiency of the PV system and ensure optimal performance, thereby contributing to the reduction of fossil fuel dependency (Naamane et al., 2024). In 2024, Tia and colleagues developed a data-driven strategy for tracking the MPP of PV systems, which is capable of flexibly adapting to dynamic environmental changes. Utilizing deep learning and incremental adjustments, this strategy effectively mitigates the loss issues caused by perturbations in the traditional perturb and observe (P&O) method, thereby significantly enhancing the efficiency of the overall solar power generation system (Tia et al., 2024).

This research focuses on enhancing the step-up charging module in DC charging stations by introducing an innovative control strategy for electric vehicle chargers operating in discontinuous conduction mode (DCM). Characterized by high gain, low cost, and compact size, this charging strategy shows significant potential for future applications in

Table 1
Comparison of five solar MPPT algorithms.

MPPT algorithm	Complexity	Parameters	Uniform sunlight environment performance	Shaded environment performance
(Kong et al., 2024)	Complex	Required	High	High
(Wesabi et al., 2024)	Medium	Not required	High	High
(Naamane et al., 2024)	Complex	Required	High	Medium
(Tia et al., 2024)	Complex	Required	High	Medium
Proposed	Medium	Not required	High	High

solar-powered DC charging station boost modules. Additionally, the circuit and charging control method address the prevalent issues of high cost and large size in current designs. Moreover, the solar step-up power converter (SSUPC) integrates a pyramid maximum power point tracking (MPPT) algorithm. This algorithm initiates MPPT to capture the maximum power point (MPP) when the solar output power is insufficient, thereby improving the power stability of the charging system.

Table 1 presents a comparison of five solar MPPT algorithms. The proposed MPPT algorithm surpasses those referenced in the literature (Kong et al., 2024; Naamane et al., 2024; Tia et al., 2024) in terms of complexity and parameter settings. In uniform sunlight and shaded environment performances, the proposed MPPT algorithm also demonstrates superior effectiveness compared to the algorithms mentioned in references (Naamane et al., 2024; Tia et al., 2024). Thus, the superiority of the proposed MPPT algorithm is clearly evident.

The remainder of this paper is organized as follows: Section 2 describes the traditional boost converter. Section 3 presents the step-up power converter integrated with a novel high-gain control strategy in detail. Section 4 provides an in-depth discussion of the proposed pyramid MPPT algorithm and explores the integration of this MPPT algorithm with the battery charging system. Section 5 displays experimental results, and Section 6 offers conclusions and suggestions for future work.

2. Tradition boost converter circuit

A traditional boost converter is a type of DC/DC power converter aimed at increasing the output voltage. Its circuit is primarily composed of four key components: an inductor, a switch, a diode, and a capacitor.

$$V_o = \frac{V_{in}}{1 - D} \tag{1}$$

In a traditional boost converter, the relationship between the output voltage (V_o), input voltage (V_{in}), and duty cycle (D) is captured by Eq. (1). Although functional, a significant limitation of this conventional boost converter is its inability to achieve high voltage gain and sufficient output voltage. This shortcoming becomes particularly pronounced in applications requiring elevated voltage levels, such as electric vehicle battery charging, where the necessary charging voltage often reaches 480 V. To tackle these challenges, this study introduces a novel approach that integrates an innovative control strategy with a solar step-up power converter (SSUPC), specifically designed for electric vehicle chargers. This method aims to effectively address the issues of insufficient voltage gain and output voltage in traditional systems.

3. Solar step-up power converter integrated with a novel control strategy

This section introduces a solar step-up power converter and an innovative high-gain control strategy. These hardware and software

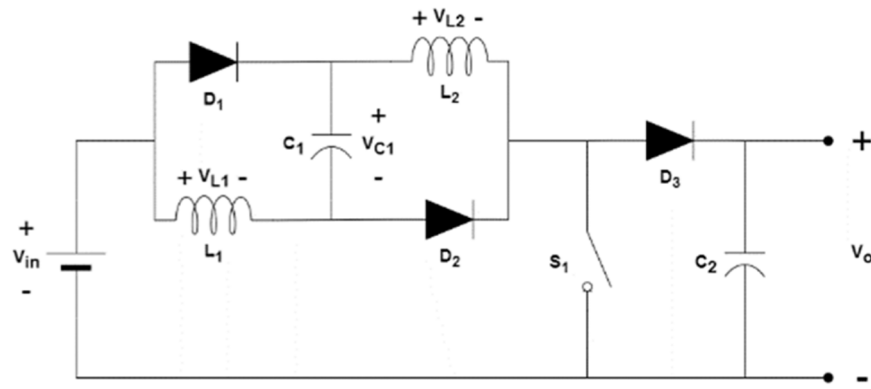
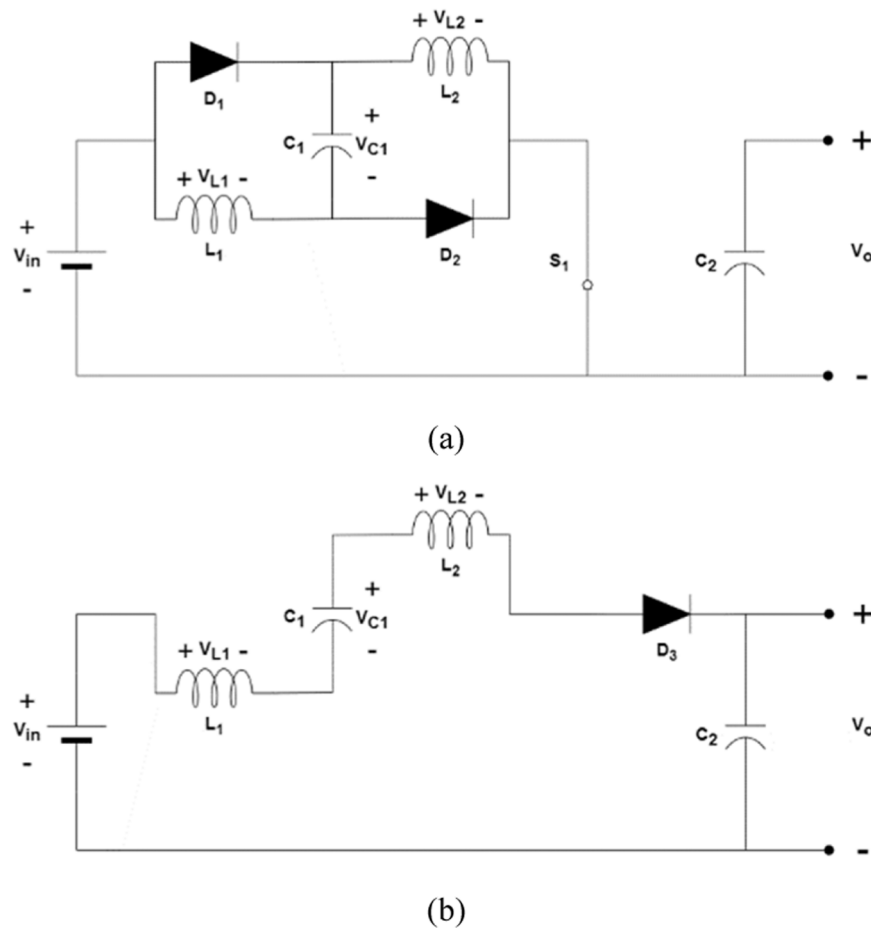


Fig. 1. Schematic diagram of the SSUPC.

Fig. 2. Operation states of the SSUPC: (a) Switch (S_1) conducting, (b) Switch (S_1) open.

designs facilitate the boosting of low voltage up to 480 volts, suitable for charging electric vehicle batteries.

3.1. Solar step-up power converter

Fig. 1 presents a schematic representation of a solar step-up power converter (SSUPC), which incorporates two inductors (L_1, L_2), two capacitors (C_1, C_2), three diodes (D_1, D_2, D_3), and a switch (S_1). Initially, when the switch S_1 is in the closed position, diode D_3 remains inactive. During this phase, the current originating from V_{in} primarily splits into two pathways: one passing through D_1, L_2 , and S_1 , and the other via L_1, D_2 , and S_1 , before reconverging at V_{in} . This configuration is equivalently

depicted in Fig. 2(a). In this mode, energy accumulates in both inductors and capacitor C_1 . Upon opening S_1 , diode D_3 becomes active, directing the current from V_{in} through L_1, C_1, L_2, D_3 , and C_2 , before it cycles back to V_{in} . At this juncture, the circuit configuration from Fig. 1 is equivalently illustrated in Fig. 2(b). In this state, the energy previously stored in the inductors and capacitors during the conducting phase of S_1 is released, effectively doubling and enhancing the output voltage.

Fig. 3 displays an SSUPC operating in continuous conduction mode (CCM), illustrating the voltage and current waveforms for inductors (L_1, L_2). In this illustration, 'D' represents the duty cycle of the switch (S_1). Within CCM, the relationship between the input voltage (V_{in}) and output voltage (V_o) can be determined using specific formulas. As shown in

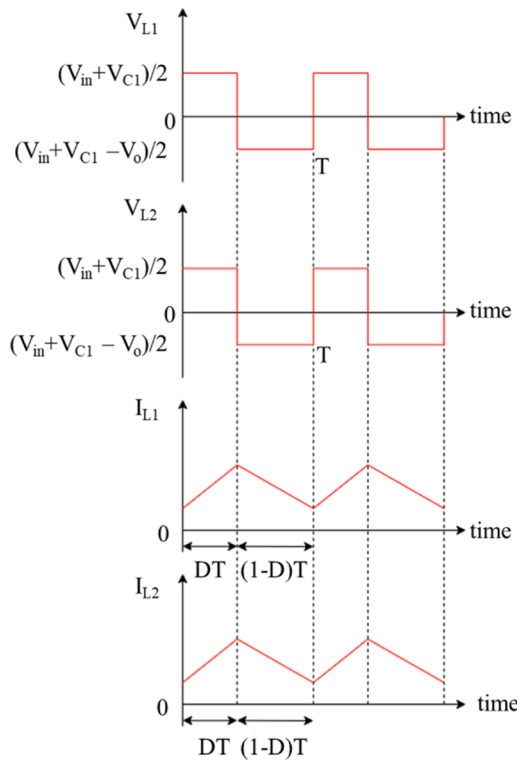


Fig. 3. Voltage and current waveforms of inductors (L_1 , L_2) in the CCM of the SSUPC.

Fig. 3, the circuit operates in a state similar to that depicted in Fig. 2(a) during the time DT , when the switch (S_1) is active. In a steady-state scenario, and by applying Kirchhoff's voltage law (KVL), the following equations can be established:

$$V_{in} - V_{L1} + V_{C1} - V_{L2} = 0 \quad (2)$$

Here, V_{L1} represents the voltage across inductor L_1 , V_{L2} represents the voltage across inductor L_2 , and V_{C1} is the voltage across capacitor C_1 .

Assuming that all components in the circuit are ideal, the corresponding expression can be formulated as follows:

$$V_{L1} = V_{C1} = V_{L2} \quad (3)$$

Upon reorganizing Eqs. (2) and (3), the derived expression can be presented as follows:

$$\frac{V_{in} + V_{C1}}{2} = V_{L1} = V_{L2} \quad (4)$$

Subsequently, the switch (S_1) transitions from the conducting state to an open state for a period of $(1-D)T$. During this phase, the circuit operates in a state resembling that depicted in Fig. 2(b). In steady-state operation, and in accordance with the KVL, the equation can be expressed as follows:

$$V_{in} - V_{L1} + V_{C1} - V_{L2} - V_o = 0 \quad (5)$$

Upon rearranging Eqs. (3) and (5), the revised expression can be presented as follows:

$$\frac{V_{in} + V_{C1} - V_o}{2} = V_{L1} = V_{L2} \quad (6)$$

Finally, since the total voltage change across the two inductors (L_1 , L_2) should be zero over one cycle, this relationship can be expressed using Eqs. (4) and (6) as follows:

$$2 \cdot \left(\frac{V_{in} + V_{C1}}{2} \right) \cdot DT + 2 \cdot \left(\frac{V_{in} + V_{C1} - V_o}{2} \right) \cdot (1-D)T = 0 \quad (7)$$

After rearranging, the relationship between the input voltage (V_{in})

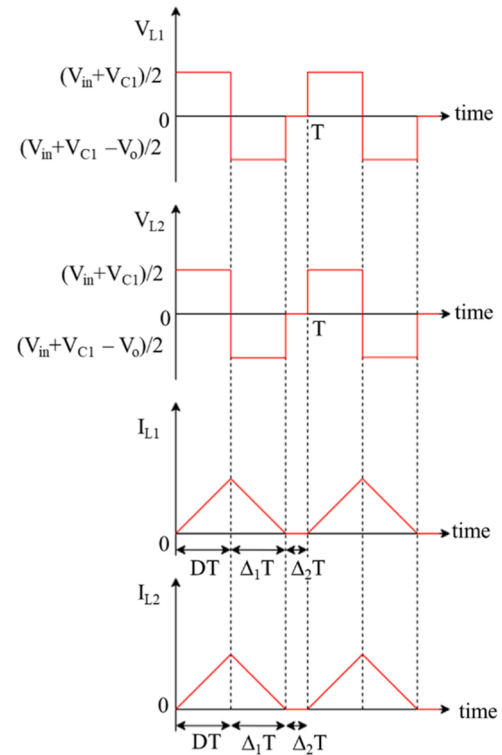


Fig. 4. Voltage and current waveforms of inductors (L_1 , L_2) in the DCM of the SSUPC.

and the output voltage (V_o) for the SSUPC operating in CCM is depicted in Eq. (8).

$$V_o = \frac{2V_{in}}{1-D} \quad (8)$$

$$I_o = \frac{I_{in}(1-D)}{2} \quad (9)$$

The relationship between the input current (I_{in}) and the output current (I_o) for the SSUPC operating in CCM is depicted in Eq. (9).

Fig. 4 illustrates an SSUPC operating in discontinuous conduction mode (DCM), showcasing the voltage and current waveforms across inductors L_1 and L_2 . In DCM, 'D' denotes the duty cycle of the switch S_1 . A key characteristic distinguishing DCM from CCM is that, when the switch is open, the inductors release energy for a limited duration, Δ_1T , followed by a Δ_2T interval during which both the voltage and current across the inductors drop to zero. Assuming steady-state operation and referring to both Fig. 4 and Equation (7), the relationship between the input voltage (V_{in}) and the output voltage (V_o) in DCM can be articulated as follows:

$$2 \cdot \left(\frac{V_{in} + V_{C1}}{2} \right) \cdot DT + 2 \cdot \left(\frac{V_{in} + V_{C1} - V_o}{2} \right) \cdot \Delta_1T = 0 \quad (10)$$

The relationship between the input voltage (V_{in}) and the output voltage (V_o) for the SSUPC in discontinuous conduction mode (DCM) has been derived and is presented as shown in Eq. (11) following rearrangement.

Table 2

Measured relationships among V_{in} , D , Δ_1 , and V_o for the SSUPC.

V_{in}	D	Δ_1	V_o
15 V	0.3	0.020	480 V
20 V	0.3	0.027	480 V
25 V	0.3	0.035	480 V

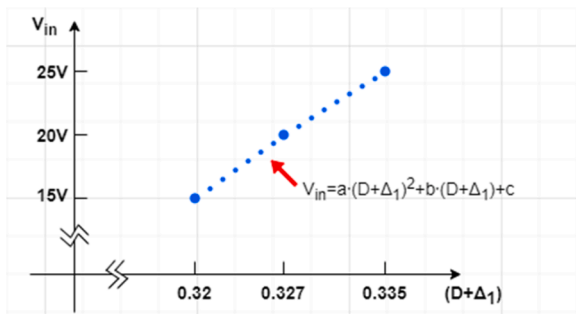


Fig. 5. Diagram illustrating the relationship between V_{in} and the variations in $D + \Delta_1$.

$$V_o = \frac{2 \cdot V_{in} \cdot (D + \Delta_1)}{\Delta_1} \quad (11)$$

3.2. Innovative high-gain control strategy

Table 2 outlines the relationship between the input voltage (V_{in}), duty cycle (D), inductor energy release interval (Δ_1), and the output voltage (V_o) for the solar step-up power converter (SSUPC) tested in this study. During the experiment, a signal generator was used to produce a pulse width modulation (PWM) signal for the switch, and a battery simulator was implemented at the load end. When V_{in} was set to 15 V and the PWM signal's duty cycle was at 0.3, the observed V_o was 480 V, Δ_1 was approximately 0.020, and the efficiency reached 93 %. With an increase in V_{in} to 20 V, while maintaining the PWM signal's duty cycle at 0.3, V_o remained at 480 V, Δ_1 was around 0.027, and the efficiency stayed near 93 %. Lastly, when V_{in} was adjusted to 25 V with the PWM signal's duty cycle constant at 0.3, V_o was 480 volts, Δ_1 approximately 0.035, and the efficiency rose to 95 %. These results demonstrate that the SSUPC in this study effectively boosts the output voltage to 480 V within an input voltage range of 15 V to 25 V. This aligns with the requirements of current electric vehicle DC charging systems. Furthermore, its conversion efficiency, exceeding 90 %, indicates minimal energy loss during the process.

In the charging system developed for this study, the microprocessor can modify the duty cycle of the PWM signal in response to variations in the input voltage (V_{in}). This adjustment is crucial for controlling the conduction duration of the switch (S_1), with the primary goal of stabilizing the output voltage within a desired range. To consistently maintain the output voltage around 480 V, in line with the requirements of electric vehicle DC charging infrastructures, we illustrate the relationship between V_{in} and the sum of D and Δ_1 . This is based on the empirical data obtained from the SSUPC, as documented in Table 2, and is graphically represented in Fig. 5. Furthermore, from this quadratic curve, we can derive Eq. (12), where the coefficients are determined to be $a = -5952.4$, $b = 4565.5$, and $c = -836.4$.

$$V_{in} = a \cdot (D + \Delta_1)^2 + b \cdot (D + \Delta_1) + c \quad (12)$$

After rearrangement, the variation of $D + \Delta_1$ as a function of changes in the input voltage (V_{in}) is succinctly expressed by Eq. (13), as follows:

$$D + \Delta_1 = \frac{-b + \sqrt{b^2 + 4a \cdot (V_{in} + c)}}{2a} \quad (13)$$

Drawing on the empirical data from the actual charging model presented in Table 2, we formulated Eq. (12). This equation enables the calculation of the sum $D + \Delta_1$ in response to variations in the input voltage (V_{in}), based on the interrelations among parameters a , b , and c . In this research, the duty cycle (D) is fixed at 0.3, thereby requiring that the value of Δ_1 remains below 0.7 to ensure operation within the discontinuous conduction mode (DCM). If the Δ_1 value, as computed by Eq. (13), exceeds 0.7, the system will no longer function effectively. This

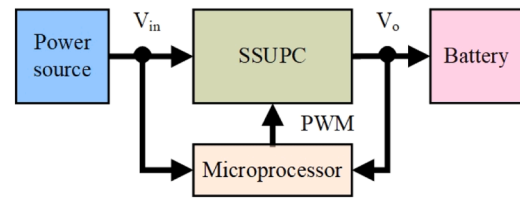


Fig. 6. Diagram of the proposed electric vehicle charging system.

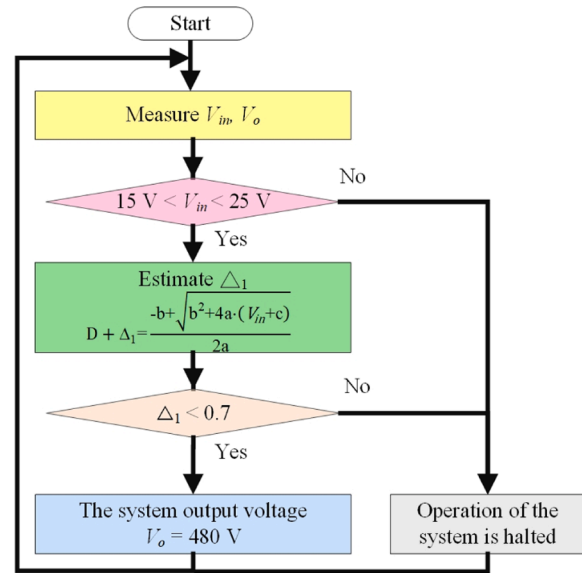


Fig. 7. Control flowchart for the novel high-gain control strategy of the proposed electric vehicle charging system.

design approach is critical to guarantee that the proposed battery charging system achieves both a high voltage gain ratio and superior efficiency.

Fig. 6 presents a schematic diagram of the proposed electric vehicle charging system, incorporating several key components: a power supply, a solar step-up power converter (SSUPC), a microprocessor, and a battery. The power supply is engineered to output a voltage that varies between 15 V and 25 V. Utilizing a novel control strategy, the microprocessor generates PWM signals to effectively drive the SSUPC. This converter, in turn, boosts the output voltage to 480 V, thereby providing the necessary power to efficiently charge the battery.

Fig. 7 presents the control flowchart for the novel high-gain control strategy of the proposed electric vehicle charging system. The process begins with the system verifying whether the input voltage is within the 15 V to 25 V range. If the voltage lies outside this range, the system will automatically shut down. If the voltage is within the specified range, the microprocessor calculates the value of Δ_1 using V_{in} and D , in accordance with Eq. (13). This calculated Δ_1 value is then fed back into the system for assessment. If Δ_1 is less than 0.7, it confirms that the charger can operate in DCM, enabling it to boost the output voltage to 480 V, which is suitable for charging electric vehicle batteries. Conversely, if Δ_1 exceeds 0.7, the system will shut down to prevent unnecessary power wastage.

4. Proposed maximum point tracking control strategy and the battery charging system

This section first introduces a newly proposed pyramid maximum power point tracking (MPPT) control strategy, which is then integrated with the proposed electric vehicle charging system to form the pyramid MPPT control strategy. In environments with abundant sunlight, the

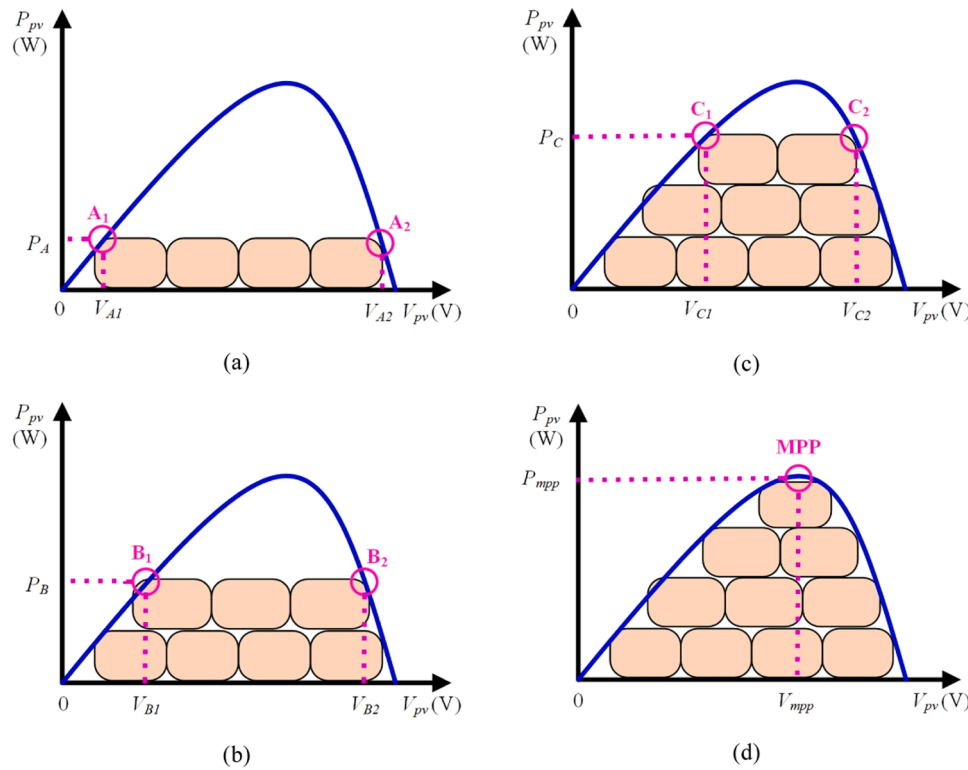


Fig. 8. Diagram illustrating the integration of PV P_{pv} - V_{pv} characteristic curves with the pyramid MPPT algorithm: (a) Establishing the pyramid base (1st layer), (b) Constructing the pyramid body (2nd layer), (c) Building the pyramid body (3rd layer), (d) Forming the pyramid apex.

solar panels can produce sufficient power; the proposed SSUPC is capable of outputting 480 V for electric vehicle charging. In situations such as early morning, evening, or when solar output is insufficient, the proposed pyramid MPPT control strategy is activated to extract the maximum power output from the solar panels for charging electric vehicles, significantly enhancing the overall efficiency of the solar charging system.

4.1. Proposed maximum point tracking control strategy

This study successfully increased low input voltages to a high output voltage of 480 V, thereby opening up new application possibilities for devices with low-voltage outputs, such as solar panels and battery packs. The research aims to enhance the application efficiency of power electronics technology in the sustainable energy sector, with a particular focus on using solar panels as the input source. Presently, there is widespread research into the application of solar power generation systems. This includes studies on solar-powered electric vehicle charging stations (Nandini et al., 2024; Huang et al., 2022), investigations into solar power systems with passive filters (Shah and Zhao, 2023), the use of maximum power point tracking (MPPT) technology (El Mezdi et al., 2023; Bishla and Khosla, 2023), and research on

solar grid-connected systems (Jaga and GhatakChoudhuri, 2023; Yaniv and Beck, 2024). These studies contribute significantly to energy conservation and carbon emission reduction, demonstrating a profound commitment to environmental protection.

In this study, we utilized a photovoltaic (PV) simulator (Chroma, 62020 H-150S) as a substitute for traditional solar panels. Operating under an irradiance of 1000 W/m^2 and at a temperature of 25°C , the simulator's technical specifications include an open-circuit voltage (V_{oc}) of 25 V, a short-circuit current (I_{sc}) of 27.7 A, a maximum power point voltage (V_{mpp}) of 20 V, a maximum power point current (I_{mpp}) of 25 A, and a maximum power point power (P_{mpp}) of 500 W. Additionally, we connected the simulator to the solar step-up power converter (SSUPC) for the purposes of electric vehicle battery charging and maximum power point tracking. Our goal was to achieve the highest possible efficiency in the solar power generation system.

Historically, researchers have developed numerous algorithms for maximum power point tracking (MPPT) technology. These include the hill-climbing (HC) method and the perturb-and-observe (P&O) method (Elgendy et al., 2015; Kjær, 2012). These techniques have proven effective in precisely identifying the maximum power point (MPP). However, they also present several challenges: (i) the operating point may oscillate around the MPP, (ii) they might not rapidly identify the MPP under changing sunlight conditions, and (iii) capturing the MPP can be difficult in shaded environments.

It's noteworthy that the SSUPC utilized in this research boasts high output voltage capabilities, obviating the need for increasing the voltage level of solar panels via series connections. Consequently, this system design can significantly ameliorate the performance issues of solar panels in shaded conditions, showcasing tremendous potential for boosting the efficiency of solar power generation. Moreover, this study seeks to advance the optimization of conventional MPPT technology, thereby introducing a novel pyramid MPPT algorithm (illustrated in Fig. 8). This innovative approach presents considerable benefits by not only preventing oscillation around the MPP but also by swiftly and

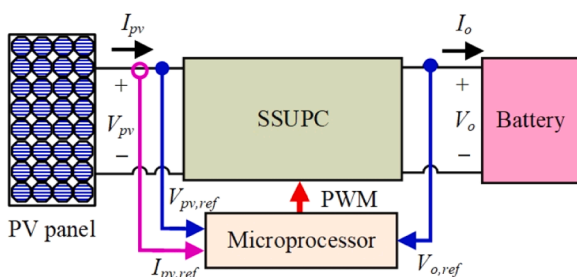


Fig. 9. Diagram of the charging control system architecture for the SSUPC.

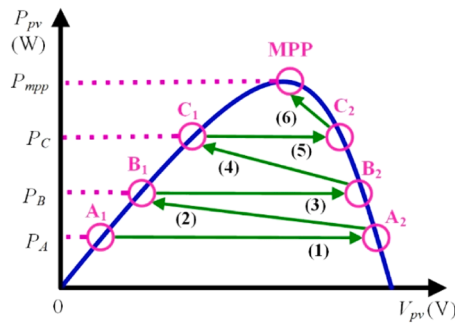


Fig. 10. Diagram illustrating the operating point of the pyramid MPPT algorithm on the PV P_{pv} - V_{pv} characteristic curve.

accurately identifying the MPP, irrespective of sunlight variability. The deployment of this algorithm can substantially improve the operational efficiency of solar power systems, guaranteeing optimal energy harvesting performance across diverse environmental scenarios.

Fig. 8 presents a diagram that illustrates the integration of the PV P_{pv} - V_{pv} characteristic curves with the pyramid MPPT algorithm. Fig. 9 depicts the architecture of the charging control system for the SSUPC employed in this study. In this study, PV panels are connected in parallel, ensuring consistent solar P_{pv} - V_{pv} characteristic curves under both uniformly lit and shaded conditions, as illustrated in Fig. 8. Fig. 8(a) demonstrates the construction of the first layer of the pyramid base. At this initial phase, the microprocessor outputs a PWM duty cycle (D_{A1}) of 0.1, as depicted in Fig. 9, situating the operating point at A_1 with a corresponding V_{pv} of V_{A1} . Through the application of Eqs. (8) and (9), it is possible to estimate P_A and P_o . Subsequently, the operating point shifts to the right half-plane in search of a duty cycle (D_{A2}) on the PV P_{pv} - V_{pv} characteristic curve that aligns with P_A and V_{A2} , a technique that facilitates the convergence of the operating point's working range. Fig. 8(b) illustrates the creation of the second layer of the pyramid. At this

juncture, the PWM duty cycle (D_{B1}) outputted by the microprocessor is calculated using Eq. (14), placing the operating point at B_1 with a V_{pv} corresponding to V_{B1} . Utilizing Equations (8) and (9), P_B and P_o are estimated. The operating point then proceeds to the right half-plane to identify a duty cycle (D_{B2}) on the PV P_{pv} - V_{pv} characteristic curve that matches P_B and V_{B2} , thereby further narrowing the operating point's work range.

$$D_{B1} = D_{A1} + k_1(D_{A2} - D_{A1}) \quad (14)$$

Wherein k_1 represents the reduction factor, D_{A1} is the duty cycle at operating point A_1 , and D_{A2} is the duty cycle at operating point A_2 .

$$D_{C1} = D_{B1} + k_2(D_{B2} - D_{B1}) \quad (15)$$

Wherein, k_2 is the reduction factor, D_{B1} is the duty cycle of operating point B_1 , and D_{B2} is the duty cycle of operating point B_2 .

Fig. 8(c) depicts the creation of the third layer of the pyramid structure. In this phase, the PWM duty cycle (D_{C1}), calculated using Eq. (15), places the operating point at C_1 , with its associated V_{pv} at V_{C1} . Utilizing Equations (8) and (9), P_C and P_o can be determined. The operating point then shifts rightward, seeking the duty cycle (D_{C2}) on the PV P_{pv} - V_{pv} characteristic curve that aligns with P_C and V_{C2} . This action moves the operating point nearer to the maximum power point (MPP). Fig. 8(d) illustrates the formation of the pyramid's apex, where the PWM duty cycle (D_{mpp}), derived from Eq. (16), positions the operating point directly at the MPP, showcasing the accurate execution of maximum power point tracking (MPPT).

$$D_{mpp} = D_{C1} + k_3(D_{C2} - D_{C1}) \quad (16)$$

Wherein, k_3 is the reduction factor, D_{C1} is the duty cycle of operating point C_1 , and D_{C2} is the duty cycle of operating point C_2 .

Fig. 10 shows the diagram illustrating the operating point of the pyramid MPPT algorithm on the PV P_{pv} - V_{pv} characteristic curve. This diagram effectively outlines the progression of the operating point from

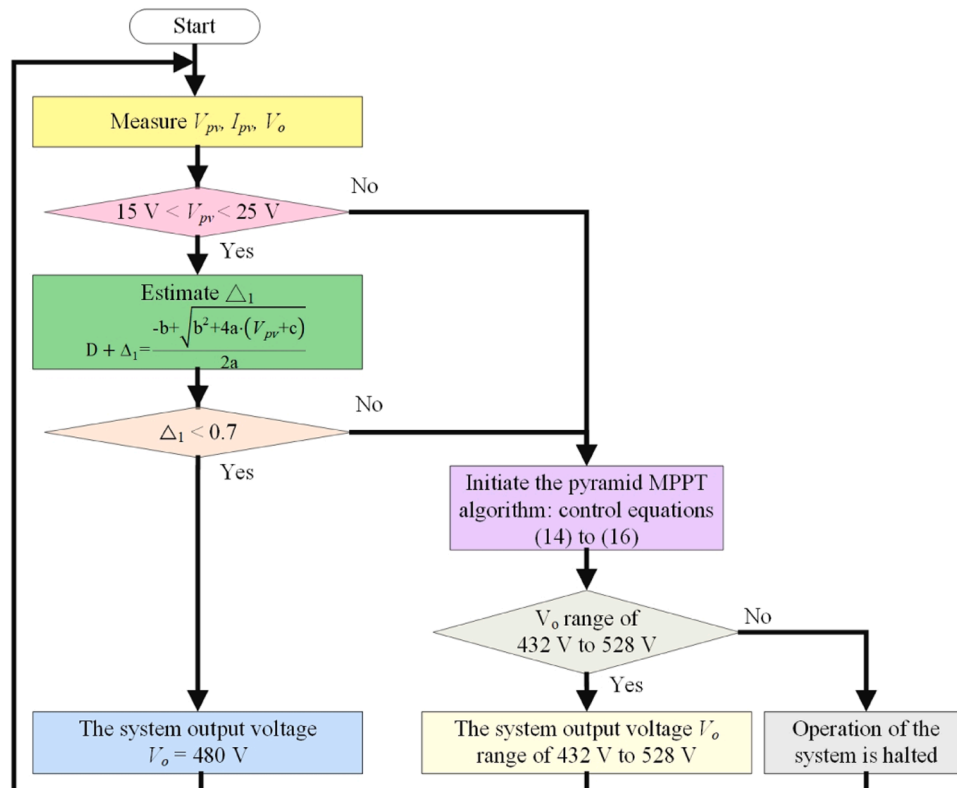


Fig. 11. Flowchart of the charging control process for the SSUPC.

Table 3
Hardware specification.

Components	Specification
Input voltage (V_{in})	15 V ~ 25 V
Output voltage (V_o)	480 V
Frequency (f_s)	8 kHz
Inductors (L_1, L_2)	45 μ H, 45 μ H
Capacitors (C_1, C_2)	100 μ F, 100 μ F, 100 μ F
Diodes (D_1, D_2, D_3)	Brand: Champion Microelectronic, P/N: PFC86G
Power MOSFET	Brand: Infineon Technologies, P/N:52N50C3
Drive IC	Brand: Toshiba, P/N: TLP250
Microprocessor	Brand: Microchip, P/N: PIC18F452
PV Simulator	Brand: Chroma, P/N: 62020 H–150S
Auxiliary power supply	Brand: GW Instek, P/N:GPS–4303
Scope	Brand: Tektronix, P/N:DPO 2014B
Battery simulator	Brand: I-Tech, P/N:IT8600

A_1 to A_2 (marking the first stage), then progresses from A_2 to B_1 (the second stage), B_1 to B_2 (the third stage), B_2 to C_1 (the fourth stage), C_1 to C_2 (the fifth stage), and finally from C_2 to the MPP (the sixth stage). These six stages clearly demonstrate that the pyramid MPPT algorithm can swiftly and accurately determine the MPP without oscillation near the MPP. Furthermore, as the intensity of sunlight changes, the PV's P_{pv} - V_{pv} characteristic curve appropriately expands or contracts. Should the operational conditions not meet the specifications defined in Eqs. (14) to (16), the operating point is reset to A_1 . This enables the MPPT to quickly adjust and more precisely capture the MPP, effectively addressing the inaccuracies faced by the hill climbing (HC) and perturb and observe (P&O) algorithms in capturing the MPP under variable lighting conditions.

4.2. The proposed control strategy for the system

Fig. 11 illustrates the flowchart of the charging control process for the SSUPC. Initially, the solar charging system utilizes the SSUPC architecture, augmented with our proposed high-gain control strategy. This setup boosts the output voltage of the solar panels from 15 V~25 V to 480 V in a discontinuous conduction mode (DCM), facilitating electric vehicle charging. However, should climatic conditions cause a decrease in PV panel output power, making it impossible for our control strategy to elevate the output voltage to 480 V, the pyramid MPPT algorithm is activated. At this juncture, the system allows an output voltage range of 432 V to 528 V. Should the output voltage maintain within this range during MPPT operation, the system persists in offering charging services to electric vehicles. In contrast, should the output voltage fall outside this designated range, the system will cease operations to safeguard both the charger and electric vehicle battery against potential damage.

5. Experimental results

Table 3 and Fig. 12 respectively provide detailed overviews of the hardware specifications and the setup configuration. The hardware includes a solar step-up power converter (SSUPC), a microprocessor, a power supply, an auxiliary power supply, a scope, a battery simulator, and other components.

Fig. 13 presents the novel high-gain control strategy applied in the proposed electric vehicle charger, featuring real-time waveforms of the input voltage (V_{in}), output voltage (V_o), and the gate-source voltage (V_{GS}) of the power MOSFET. Initially, when the microprocessor detects the input voltage (V_{in}) and the output voltage (V_o) as shown in Fig. 6, it proceeds to compute using the proposed high-gain control strategy, as

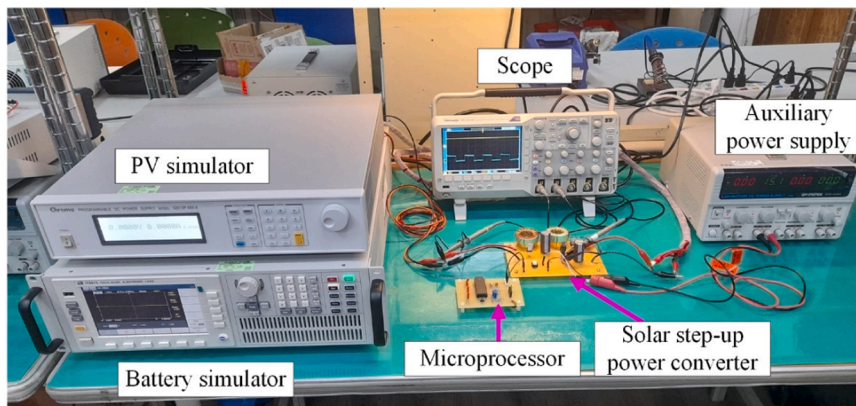


Fig. 12. The hardware testing platform for this research.

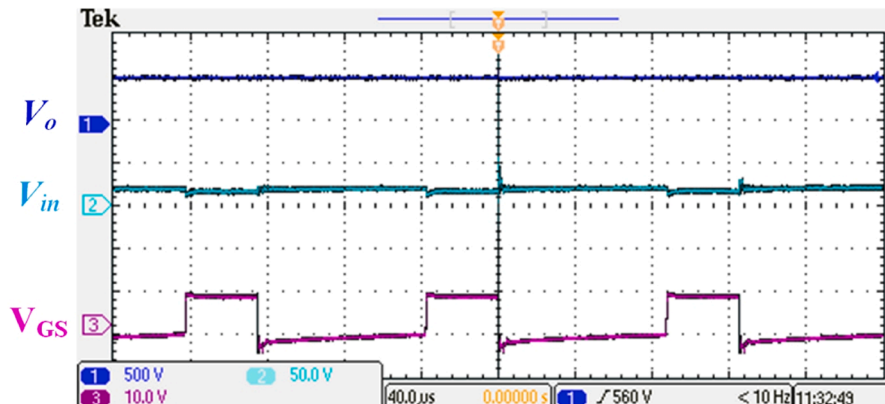


Fig. 13. Experimental waveforms of V_o , V_{in} , and V_{GS} . (V_o : 500 V/div, V_{in} : 50 V/div, V_{GS} : 10 V/div; Horizontal scale: 40 μ s/div).

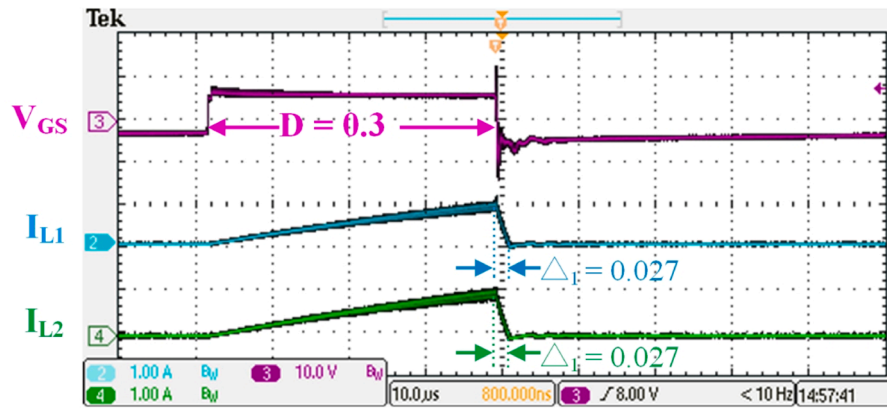


Fig. 14. Experimental waveforms of V_{GS} , I_{L1} , and I_{L2} . (V_{GS} : 10 V/div, I_{L1} : 1 A/div, I_{L2} : 1 A/div; Horizontal scale: 10 μ s/div).

Table 4

The system real test results under different conditions.

V_{in}	D	Δ_1	V_o	Efficiency
15 V	0.3	0.020	480 V	93 %
18 V	0.3	0.024	480 V	93 %
20 V	0.3	0.027	480 V	93 %
22 V	0.3	0.030	480 V	94 %
25 V	0.3	0.035	480 V	95 %

illustrated in Fig. 7. At this point, the microprocessor outputs a PWM signal with a duty cycle of 0.3 to control the power MOSFET (S_1) in the solar step-up power converter (SSUPC). With V_{in} set at 20 V and V_{GS} at 0.3, the system effectively boosts V_o to approximately 480 V, meeting the voltage requirements for DC charging systems in electric vehicles.

Fig. 14 demonstrates the innovative high-gain control strategy used in the proposed electric vehicle charger, with actual measurements of V_{GS} and the currents through inductors L_1 and L_2 (I_{L1} and I_{L2}). Under conditions where the input voltage is 20 V and V_{GS} is 0.3, the output voltage reaches 480 V, as shown in Fig. 13. Further measurements of I_{L1} and I_{L2} in Fig. 14 clearly indicate that I_{L1} and I_{L2} operate in discontinuous conduction mode (DCM), as illustrated in Fig. 4. It was also found that Δ_1 is 0.027, in accordance with Eq. (13). Therefore, this innovative high-gain control strategy focuses on high-voltage and low-current output, not only achieving more effective high-voltage gain but also outperforming continuous conduction mode (CCM).

Table 4 showcases the actual test results of the system under various conditions. With an input voltage (V_{in}) ranging from 15 V to 25 V, a fixed duty cycle (D) of 0.3, and a Δ_1 value fluctuating between 0.020 and 0.035, the system consistently produces an output voltage (V_o) of 480 V

and a rated power output of 500 W. Moreover, the system's efficiency consistently falls between 93 % and 95 %. These experimental results clearly demonstrate the system's high performance and effectiveness.

Fig. 15 displays the experimental waveforms of V_{pv} , I_{pv} , and P_{pv} for the SSUPC, utilizing the pyramid MPPT algorithm under an irradiance of 1000 W/m² and a temperature of 25 °C. Under ideal conditions, the MPP parameters are V_{mpp} at 20 V, I_{mpp} at 25 A, and P_{mpp} at 500 W. Initially, at time t_a , the proposed pyramid MPPT algorithm is activated and searches for the MPP using Eqs. (14) to (16), as shown in Fig. 10. Subsequently, at time t_b , the pyramid MPPT algorithm captures the MPP, at which point the actual power output of the PV panel (P_{pv}) reaches 496 W, thereby achieving an MPPT efficiency of 99 % (as shown in Table 5).

Fig. 16 showcases the experimental waveforms of V_{pv} , I_{pv} , and P_{pv} associated with the SSUPC employing the pyramid MPPT algorithm, under conditions of 500 W/m² irradiance and a temperature of 25 °C. Under ideal conditions, the MPP parameters include V_{mpp} at 19.2 V, I_{mpp} at 13 A, and P_{mpp} at 250 W. Initially, at time t_a , the proposed pyramid MPPT algorithm is activated and searches for the MPP using Eqs. (14) to (16), as detailed in Fig. 10. Subsequently, at time t_b , the pyramid MPPT algorithm successfully captures the MPP, at which the actual power output of the PV panel (P_{pv}) reaches 248 W, thereby achieving an MPPT

Table 5

Pyramid MPPT algorithm measured results.

Irradiance level	Temperature	P_{mpp}	P_{pv}	MPPT Efficiency
1000 W/m ²	25 °C	500 W	496 W	99 %
500 W/m ²	25 °C	250 W	248 W	99 %

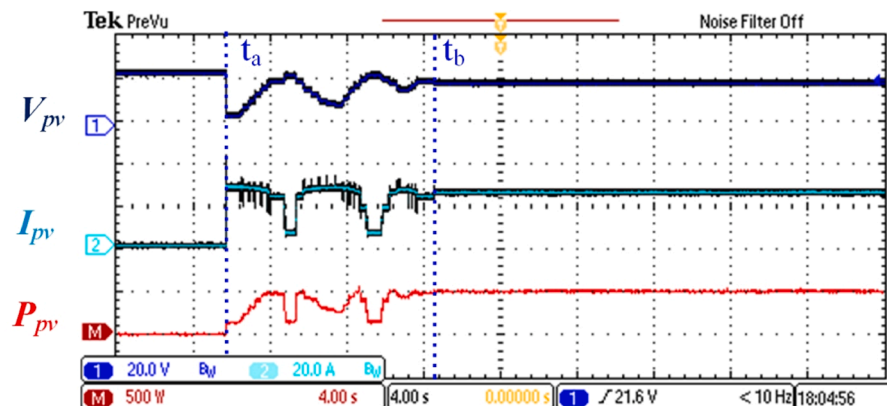


Fig. 15. Experimental waveforms of V_{pv} , I_{pv} , and P_{pv} associated with the SSUPC utilizing the pyramid MPPT algorithm, recorded at an irradiance of 1000 W/m² and a temperature of 25 °C. (V_{pv} : 20 V/div, I_{pv} : 20 A/div, P_{pv} : 500 W/div; Horizontal scale: 4 s/div).

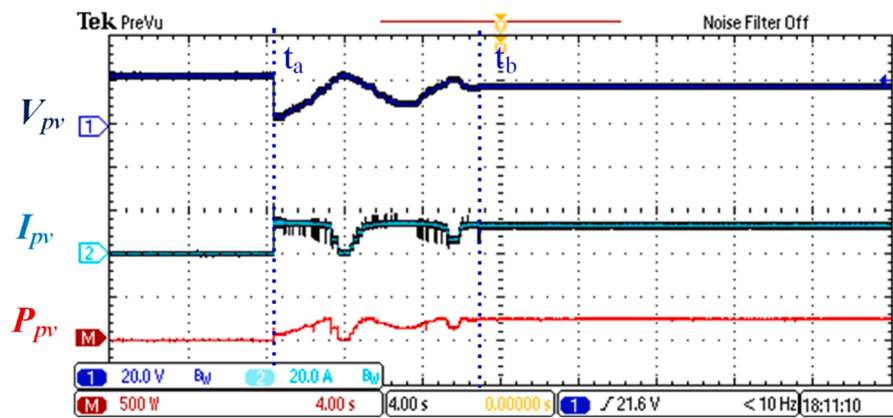


Fig. 16. Experimental waveforms of V_{pv} , I_{pv} , and P_{pv} associated with the SSUPC utilizing the pyramid MPPT algorithm, recorded at an irradiance of 500 W/m^2 and a temperature of 25°C . (V_{pv} : 20 V/div , I_{pv} : 20 A/div , P_{pv} : 500 W/div ; Horizontal scale: 4 s/div).

efficiency of 99 % (as shown in Table 5).

6. Conclusions

The innovative battery charging control strategy introduced in this study revolutionizes the DC charging process for electric vehicle batteries. A standout feature of this system is the voltage gain of the solar step-up power converter (SSUPC), which is twice that of conventional DC boost converters. Experimentally, the microprocessor in this system can control the power switch by inputting the appropriate duty cycle ratio of PWM signals, in accordance with the innovative high-gain charging control strategy proposed. The SSUPC has proven its ability to efficiently boost input voltages ranging from 15 V to 25 V up to a substantial 480 V , while maintaining an impressive conversion efficiency of between 93 % and 95 %. Furthermore, this research integrates the pyramid MPPT algorithm with the SSUPC to enhance the efficacy of solar chargers. Experiments performed at irradiance levels of 1000 W/m^2 and 500 W/m^2 have shown that MPPT efficiency can reach up to 99 %. Ultimately, the system architecture proposed in this research addresses performance degradation in solar power systems under shaded conditions. Thus, the system developed through this study is positioned as an efficacious solution for next-generation solar-powered electric vehicle chargers.

Additionally, the proposed control strategy is straightforward and easy to implement, making it suitable for large-scale production. When compared to existing electric vehicle DC charging boost modules, the system developed in this study not only offers advantages such as higher efficiency, smaller size, simpler structure, and lower cost, but also holds potential for widespread application in the boost modules of electric vehicle DC charging stations in the future. This approach would lead to a more streamlined design, thereby better meeting the requirements of economic efficiency.

CRedit authorship contribution statement

Hwa-Dong Liu: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Wen-Tung Huang:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Acknowledgements

The authors gratefully acknowledge the financial support of the National Science and Technology Council, Taiwan, grant number MOST 112–2221-E-003–003, the NTUS Innovation Cooperation 11312111001, and the National Taiwan Normal University (NTNU), Taiwan.

References

- Bishla, Sandeep, Khosla, Anita, 2023. Enhanced chimp optimized self-tuned FOPR controller for battery scheduling using Grid and Solar PV Sources. *J. Energy Storage* vol. 66, 107403. <https://doi.org/10.1016/j.est.2023.107403> (Aug.).
- Chen, J., Bei, G., Zhang, Q., Huang, L., Liu, K., Chen, P., 2023. Integrated reactive power optimization for distribution systems considering electric vehicle dis-/charging support. *Energy Rep.* vol. 9, 1888–1896. <https://doi.org/10.1016/j.egy.2023.04.189>.
- Deshmukh, S., Iqbal, A., Islam, S., Khan, I., Marzband, M., Rahman, S., Al-Wahedi, A.M. A.B., 2022. Review on classification of resonant converters for electric vehicle application. *Energy Rep.* vol. 8, 1091–1113. <https://doi.org/10.1016/j.egy.2021.12.013>.
- El Mezdi, Karim, El Magri, Abdelmounime, Watil, Aziz, El Myasse, Ilyass, Bahatti, Lhoucine, Lajouad, Rachid, Ouabi, Hassan, 2023. Nonlinear control design and stability analysis of hybrid grid-connected photovoltaic-Battery energy storage system with ANN-MPPT method. *J. Energy Storage* vol. 72 (Part E), 108747. <https://doi.org/10.1016/j.est.2023.108747> (Nov.).
- Elgendy, M.A., Zahawi, B., Atkinson, D.J., 2015. Operating characteristics of the P&O algorithm at high perturbation frequencies for standalone PV systems. *IEEE Trans. Energy Convers.* vol. 30 (1), 189–198. <https://doi.org/10.1109/TEC.2014.2331391> (Mar.).
- Emamalipour, R., Lam, J., 2020. A hybrid string-inverter/rectifier soft-switched bidirectional DC/DC converter. *IEEE Trans. Power Electron.* vol. 35 (8), 8200–8214. <https://doi.org/10.1109/TPEL.2019.2962518> (Aug.).
- Hakam, Y., Gaga, A., Elhadadi, B., 2024. Exploring the state of electric vehicles: an evidence-based examination of current and future electric vehicle technologies and smart charging stations. *Energy Rep.* vol. 11, 4102–4114. <https://doi.org/10.1016/j.egy.2024.04.002>.
- Huang, Pei, Tu, Ran, Zhang, Xingxing, Han, Mengjie, Sun, Yongjun, Hussain, Syed Asad, Zhang, Linfeng, 2022. Investigation of electric vehicle smart charging characteristics on the power regulation performance in solar powered building communities and battery degradation in Sweden. *J. Energy Storage* vol. 56 (Part A), 105907. <https://doi.org/10.1016/j.est.2022.105907> (Dec.).
- Jaga, Om. Prakash, GhatakChoudhuri, Sumit, Nov. 2023. A self-reliant, smart control scheme for grid-tied, solar energy conversion-battery energy storage system. *J. Energy Storage* vol. 72 (Part A), 108327. <https://doi.org/10.1016/j.est.2023.108327>.
- Karneddi, H., Ronanki, D., 2022. Universal bridgeless nonisolated battery charger with wide-output voltage range. *IEEE Trans. Power Electron.* vol. 11 (3), 2816–2820. <https://doi.org/10.1109/TPEL.2022.3217943>.
- Kjær, S.B., 2012. Evaluation of the “Hill Climbing” and the “Incremental Conductance” maximum power point trackers for photovoltaic power systems. *IEEE Trans. Energy Convers.* vol. 27 (4), 922–929. <https://doi.org/10.1109/TEC.2012.2218816> (Dec.).
- Kong, L., Wang, B., Fan, D., Shi, S., Ouyang, X., Xu, M., 2024. Optimize photovoltaic MPPT with improved snake algorithm. *Energy Rep.* vol. 11, 5033–5045. <https://doi.org/10.1016/j.egy.2024.04.064>.

- Kumar, B., Anil, Jyothi, B., Rathore, R.S., Singh, A.R., Kumar, B.H., Bajaj, M., 2023. A novel framework for enhancing the power quality of electrical vehicle battery charging based on a modified Ferdowsi converter. *Energy Rep.* vol. 10, 2394–2416. <https://doi.org/10.1016/j.egy.2023.09.070>.
- Lenka, R.K., Panda, A.K., Dash, A.R., Senapati, L., Tiwary, N., 2023. A unified control of grid-interactive off-board EV battery charger with improved power quality. *IEEE Trans. Transp. Electr.* vol. 9 (1), 920–933. <https://doi.org/10.1109/TTE.2022.3172354> (Mar.).
- Li, S.E., Guo, Q., Xin, L., Cheng, B., Li, K., 2017. Fuel-saving servo-loop control for an adaptive cruise control system of road vehicles with step-gear transmission. *IEEE Trans. Veh. Technol.* vol. 66 (3), 2033–2043. <https://doi.org/10.1109/TVT.2016.2574740> (Mar.).
- A.A. Mamun, S. Arfin, S. Khan, “High Gain DC-DC Converter for Three-Wheeler Electric Vehicles,” 2021 IEEE International IOT, Electronics and Mechatronics Conference, Apr. 2021, doi: 10.1109/IEMTRONICS52119.2021.9422579.
- Naamane, D., Benbouhenni, H., Chebabhi, A., Laid, Z., Zellouma, D., Colak, I., 2024. A new nonlinear control to improve the efficiency of the PV-SAPF system. *Energy Rep.* vol. 11, 3096–3116. <https://doi.org/10.1016/j.egy.2024.01.020>.
- Nandini, K.K., Jayalakshmi, N.S., Kumar Jadoun, Vinay, 2024. A combined approach to evaluate power quality and grid dependency by solar photovoltaic based electric vehicle charging station using hybrid optimization. *J. Energy Storage* vol. 84 (Part B), 110967. <https://doi.org/10.1016/j.est.2024.110967> (Apr.).
- Nazi, H., Babaei, E., Tohidi, S., Liserre, M., 2023. An isolated SRC-based single phase single stage battery charger for electric vehicles. *IEEE Trans. Transp. Electr.* vol. 9 (1), 1252–1262. <https://doi.org/10.1109/TTE.2022.3185018> (Mar.).
- Payarou, T., Pillay, P., 2023. Integrated multipurpose power electronics interface for electric vehicles. *IEEE Trans. Transp. Electr.* vol. 9 (2), 2429–2443. <https://doi.org/10.1109/TTE.2022.3209098> (Jun.).
- Rafi, M.A.H., Bauman, J., 2023. High-efficiency variable turns-ratio semi-dual active bridge converter for a DC fast charging station with energy storage. *IEEE Trans. Transp. Electr.* vol. 9 (3), 3918–3935. <https://doi.org/10.1109/TTE.2022.3231554> (Sep.).
- Ramanathan, G., Bharatiraja, C., Kotb, H., Emara, A., 2024. PV-assisted modified Z-source inverter for multiport EV charging infrastructure: access with PV2V. *Energy Rep.* vol. 11, 5716–5732. <https://doi.org/10.1016/j.egy.2024.05.036>.
- Rathore, V., Reddy, S.R.P., Rajashekara, K., 2022. An isolated multilevel DC–DC converter topology with hybrid resonant switching for EV fast charging application. *IEEE Trans. Ind. Appl.* vol. 58 (5), 5546–5557. <https://doi.org/10.1109/TIA.2022.3168504>.
- Sayed, S.S., Massoud, A.M., 2022. Review on state-of-the-art unidirectional non-isolated power factor correction converters for short-/long-distance electric vehicles. *IEEE Access* vol. 10, 11308–11340. <https://doi.org/10.1109/ACCESS.2022.3146410> (Jan.).
- Shah, P., Zhao, X., 2023. Leakage current mitigation technique in solar PV array system using passive filter. *IEEE Trans. Energy Convers.* vol. 38 (1), 463–478. <https://doi.org/10.1109/TEC.2022.3186741> (Mar.).
- Shahverdi, M., Jamehbozorg, A., Serrato, C., Flores, N., 2023. Learning-based hierarchical control for a net-zero commercial building with solar plus storage and high EV Chargers Penetration. *Energy Rep.* vol. 10, 3724–3732. <https://doi.org/10.1016/j.egy.2023.10.032>.
- Sharma, U., Singh, B., 2023a. A bidirectional onboard charger with multistep constant current charging capability. *IEEE Trans. Transp. Electr.* vol. 9 (1), 1227–1237. <https://doi.org/10.1109/TTE.2022.3184527> (Mar.).
- Sharma, U., Singh, B., 2023b. A bidirectional charger for low-voltage-powered battery vehicles. *IEEE Trans. Transp. Electr.* vol. 9 (3), 3994–4003. <https://doi.org/10.1109/TTE.2023.3236965> (Sep.).
- Thangavel, S., Mohanraj, D., Girijaprasanna, T., Raju, S., Dhanamjayulu, C., Mueyen, S. M., 2023. A comprehensive review on electric vehicle: battery management system, charging station, traction motors. *IEEE Access* vol. 11, 20994–21019. <https://doi.org/10.1109/ACCESS.2023.3250221> (Feb.).
- Tia, L., Dong, C., Mu, Y., Jia, H., 2024. DPGS: Data-driven photovoltaic grid-connected system exploiting deep learning and two-stage single-phase inverter. *Energy Rep.* vol. 11, 1910–1924. <https://doi.org/10.1016/j.egy.2024.01.038>.
- Wang, W.V., Thrimawithana, D.J., 2020. A novel converter topology for a primary-side controlled wireless EV charger with a wide operation range. *IEEE J. Emerg. Sel. Top. Ind. Electron.* vol. 1 (1), 36–45. <https://doi.org/10.1109/JESTIE.2020.3003357> (Jul.).
- Wesabi, I.A., Fang, Z., Farh, H.M.H., A-Shamma'a, A.A., Shaalan, A.M.A., 2024. Comprehensive comparisons of improved incremental conductance with the state-of-the-art MPPT Techniques for extracting global peak and regulating dc-link voltage. *Energy Rep.* vol. 11, 1590–1610. <https://doi.org/10.1016/j.egy.2024.01.020>.
- Yaniv, Arbel, Beck, Yuval, 2024. Optimal battery scheduling in solar-plus-storage grid-connected microgrid for profit and cost efficiency: a use case on an Israeli microgrid. *J. Energy Storage* vol. 77, 109697. <https://doi.org/10.1016/j.est.2023.109697> (Jan.).
- Zheng, P., Bauman, J., 2023. High efficiency bidirectional LLC+C resonant converter with parallel transformers for solar-charged electric vehicles. *IEEE Trans. Transp. Electr.* vol. 9 (1), 1428–1442. <https://doi.org/10.1109/TTE.2022.3199157> (Mar.).